SEQUENCE 12

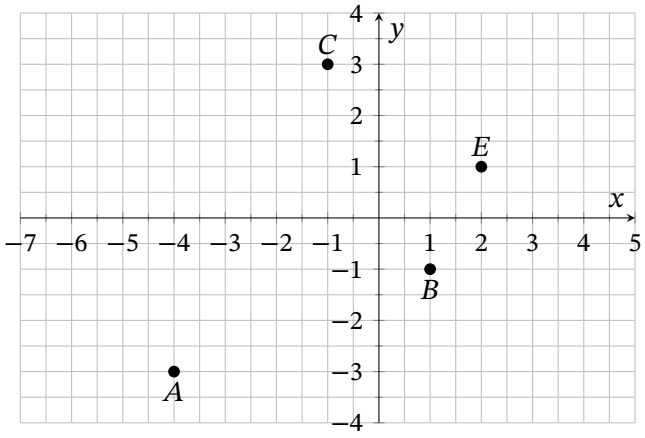
Coordinate geometry: vectors and lines synthesis

15–26 March 2027

12.1 Parallelogram and intersection

Example 1 · Booklet synthesis. In an orthonormal frame, $A(-4,-3)$, $B(1,-1)$ and $C(-1,3)$.

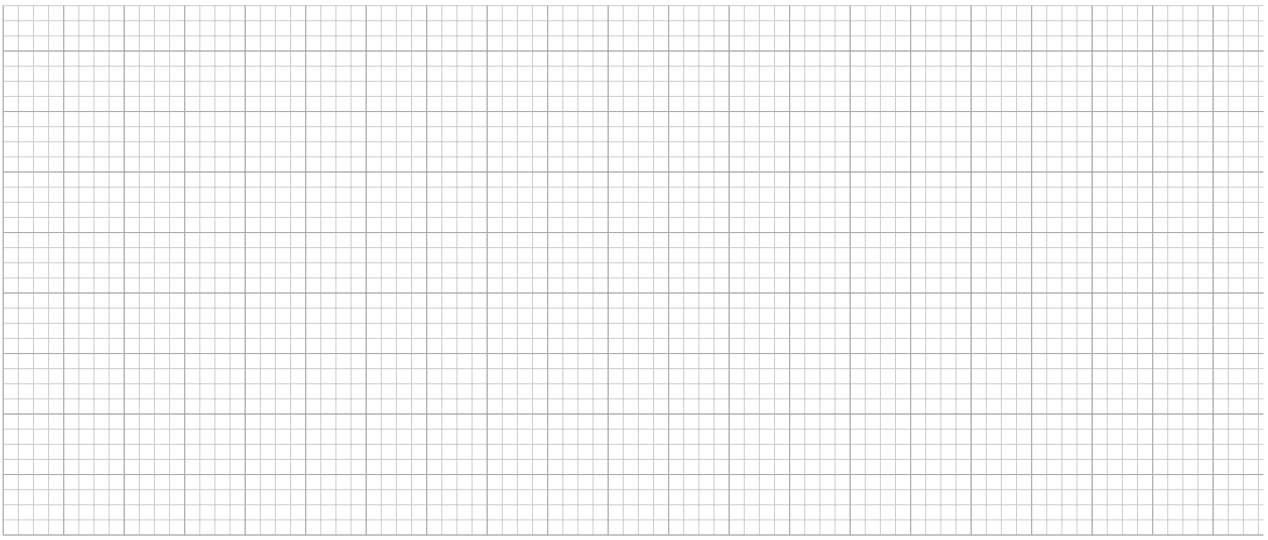
1. Determine D such that $ABCD$ is a parallelogram.
2. Determine its centre I in two ways.
3. With $E(2,1)$, determine K, the intersection of (AB) and (CE).
4. Determine F such that $\overrightarrow{AF} = \frac{2}{3}\overrightarrow{AE}$. Are D, F, B collinear?



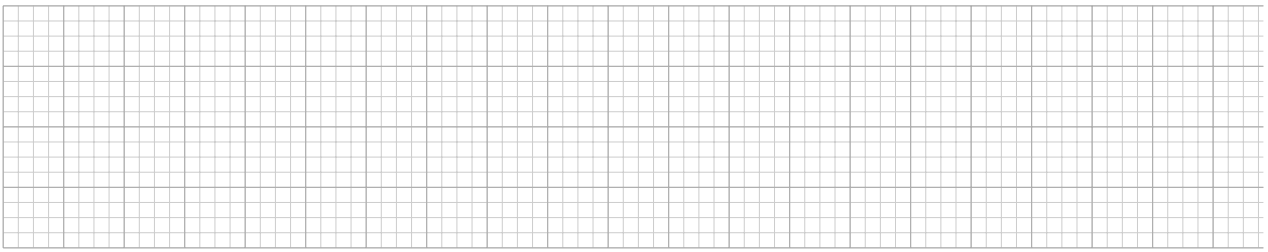
12.2 Comparing a vector method and a coordinate method

Example 2. In a non-degenerate triangle ABC, I is the midpoint of $[AB]$ and J the midpoint of $[AC]$.

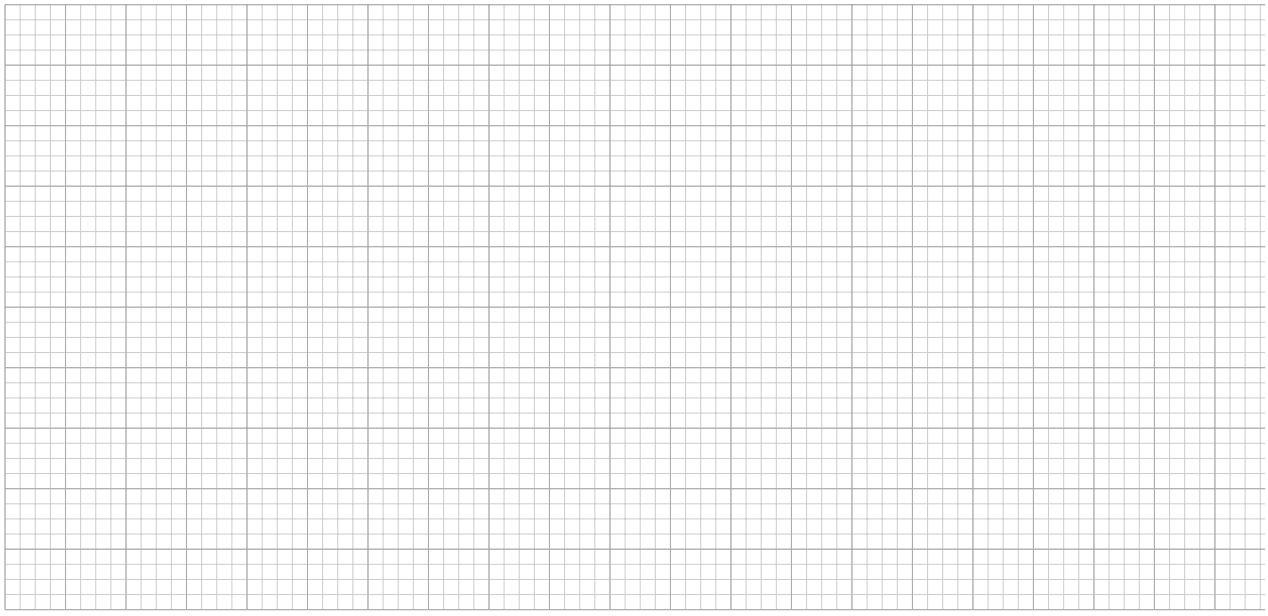
1. Prove with vectors that $(IJ) \parallel (BC)$.
2. Check with $A(0,0)$, $B(6,0)$, $C(2,4)$.
3. For these coordinates, determine the equations of (IJ) and (BC), then the ratio IJ/BC.



Example 3. A figure seems to show two parallel lines. What is the difference between this observation and a proof?



12.3 Recognising a configuration and justifying

Example 4 · Original situation. In an orthonormal frame, $A(0,0)$, $B(4,2)$, $C(3,4)$ and $D(-1,2)$.

1. Show that $ABCD$ is a parallelogram.
2. Calculate the four lengths and determine whether it is a rhombus.
3. Calculate AC^2, BD^2 and determine whether it is a rectangle.
4. Determine its centre and an equation of each diagonal.